OpCode-Stats: A Statistical Analysis of Compiled Program Instruction Sequences

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Abstract

Compiled binary executables, when viewed as flat sequences of machine opcodes, exhibit rich statistical structure that shares properties observed in natural language and biological sequences. We present *OpCode-Stats*[1], an open-source toolkit (inspired by computational biology techniques applied to DNA sequences) that applies computational biology and information-theoretic techniques to the systematic study of program instruction sequences. Across multiple corpora, three independent validation corpora, and a 100-binary compiler matrix, we characterise opcode frequency distributions, n-gram entropy rates, compressibility, recurring motifs, and manifold dimensionality. Systematic validation shows that the core findings—entropy structure, compressibility, motif recurrence, and low-dimensional manifold occupancy—persist after aggressive boilerplate stripping (max 3.2% change), replicate across independent corpora, and are compiler-invariant (compiler $R^2 < 0.03$). The toolkit and all experimental data are publicly available.

1 Introduction

The last decade has witnessed remarkable progress in language models trained on source code [1, 3]. Yet the statistical properties that make code so learnable remain poorly characterised at the machine-instruction level. Source code is parsed into abstract syntax trees, compiled through multiple transformation passes, and ultimately reduced to a sequence of opcodes—a representation that discards all high-level structure yet, as we show here, retains a profound statistical regularity. Bioinformatics offers a natural set of tools for studying such sequences. DNA and protein sequences are studied with k-mer analysis, entropy profiling, motif discovery, and information-theoretic metrics [18]. We propose that binary instruction sequences—hereafter *opcode sequences*—are amenable to the same treatment.

Contributions. This paper makes the following contributions:

1. A reproducible, open-source analysis toolkit (*OpCode-Stats*, https://github.com/OpCode-Stats/OpCode-Stats) that applies computational biology techniques to compiled binaries.
2. Empirical characterisation of opcode frequency concentration: a heavy- tailed rank-frequency distribution (MLE $\hat{\alpha} \approx 1.4$; KS $= 0.117$) with a unigram–sequential decomposition of compression gains showing that 69% of compressibility advantage over random sequences comes from frequency skew and 31% from sequential structure.
3. Quantification of n-gram entropy structure and its deviation from unigram-shuffled baselines, revealing multi-scale sequential dependencies.
4. Systematic motif discovery linking recurrent k-mers to specific compiler-generated idioms.
5. A thin-manifold characterisation of the program space via compression, mutual information decay, and corpus-level PCA.
6. A controlled compiler matrix experiment (5 projects $\times$ {gcc, clang} $\times$ {`-O0`, `-O2`, `-O3`}) showing that both source-code identity and compiler identity are comparably strong determinants of

[1] https://github.com/OpCode-Stats/OpCode-Stats

binary similarity (both $p < 10^{-4}$, $r \approx 0.5$; project vs. compiler not significantly different, $p = 0.38$) and that compiler variation adds new manifold dimensions (d_{95} doubles from 8 to 17 when compiler axis is included).

7. Empirical validation that n-gram language model perplexity closely tracks the empirical conditional entropy (bigram gap < 0.07 bits), closing the loop between statistical structure and model performance.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 describes our pipeline. Section 4 presents empirical findings. Section 5 presents validation and robustness checks. Section 6 interprets results in the context of language model performance. Section 7 concludes.

2 Background and Related Work

2.1 Opcode-Level Program Analysis

Binary analysis for security and malware detection has a long history of exploiting opcode frequency features [2, 13]. Bilar [2] observed Zipf-like distributions in Windows PE files, noting that a small set of opcodes accounts for most instruction traffic. Subsequent work used n-gram opcode features for malware classification [13, 15], but focused on discriminative accuracy rather than the underlying information structure.

The closest prior cross-domain study is Gan et al. [7], who applied n-tuple Zipf analysis to DNA, English text, source code, and binary files. Our toolkit extends this comparison with a broader suite of information-theoretic measures (mutual information decay, LZ78 complexity, motif discovery, clone detection) and a controlled compiler-variation experiment. In the malware domain, MAlign [14] encodes raw binary bytes as nucleotides and applies genome sequence alignment to malware clustering, demonstrating the practical value of the biological sequence analogy for binary analysis.

Recent binary code embedding systems implicitly validate the existence of statistical structure in assembly but do not provide direct characterisation. PalmTree [11] learns BERT-style instruction embeddings from assembly corpora; jTrans [19] extends this with jump-aware Transformers for binary similarity detection. Both implicitly rely on the statistical regularities that OpCode-Stats makes explicit and measurable. Marcelli et al. [12] provide the most comprehensive systematisation of binary function similarity, evaluating approaches across 243K binaries and 26.8M functions; their benchmark is the natural target for future large-scale validation of OpCode-Stats's similarity modules. HermesSim [20] argues that graph-based semantic representations outperform NLP-style n-gram approaches for binary *similarity search*; this is orthogonal to our focus on statistical characterisation rather than similarity ranking. Nova [9], a generative language model for assembly, similarly demonstrates learnability but not measurement.

2.2 Compression-Based Similarity

Normalized Compression Distance (NCD) [4] provides a parameter-free measure of similarity grounded in Kolmogorov complexity. Its approximation via `zlib` or `lzma` has been applied to clustering music, genomes, and source files [4]. We apply NCD systematically to binary instruction sequences and compare it against n-gram TF-IDF similarity.

2.3 Information Theory of Programs

Shannon entropy and mutual information have been applied to source code to measure predictability [8, 17]. The *naturalness* hypothesis [8] states that software is more predictable than natural language because it is written by humans following conventions; cross-entropy of token-level language models on code is lower than on English prose. Our work extends this line to the binary level, measuring structural properties that survive compilation.

2.4 Motif Discovery in Sequences

K-mer based motif discovery originated in genomics for finding transcription factor binding sites [5]. In the binary domain, compiler idiom recognition has been studied in the context of decompilation and binary lifting [16], but systematic statistical characterisation of recurrent opcode patterns is absent from the literature.

3 Methodology

3.1 Corpus Design and Extraction

We analyse two corpora of GNU/Linux ELF x86-64 binaries drawn from `/usr/bin`:

- *Smoke* (15 binaries): grep, sed, gzip, bzip2, xz, find, tar, bash, python3, curl, git, vim, nano, gcc, objdump. Verified to have distinct inodes, eliminating hard-link duplicates.

- *Coreutils-system* (40 binaries after inode deduplication): drawn from 11 functional categories (text processing, compression, filesystem, networking, shells, editors, compilers, interpreters, version control, archives, utilities) with a 10 MB size cap. Adds awk, diff, zip, unzip, cp, mv, rm, chmod, chown, df, wget, ssh, scp, nc, dash, clang, as, ld, perl, tar, ar, strace, nm, readelf, addr2line to the smoke set.

Instruction sequences are extracted via `objdump -d` with a 60-second timeout per binary. Only the mnemonic of each instruction is retained; operands and addresses are discarded. This reduces each binary to a flat opcode sequence $s = (o_1, o_2, \ldots, o_N)$ where o_i belongs to a shared vocabulary $\mathcal{V}$. Functions are extracted separately to support per-function motif and positional analyses.

3.2 Frequency and Zipf Analysis

Let $f(v)$ denote the corpus-wide frequency of opcode $v \in \mathcal{V}$ and $r(v)$ its frequency rank. We fit Zipf's power law

$$f(v) \;\propto\; r(v)^{-\alpha} \tag{1}$$

via OLS on log-log coordinates. A 95% bootstrap confidence interval on α is obtained from 200 resamplings with replacement. As a complementary estimator, we also apply maximum likelihood estimation (MLE) to the truncated discrete Zipf model: we maximise $\ell(\alpha) = -\alpha \sum_r f(r) \log r - N \log Z_V(\alpha)$ where $Z_V(\alpha) = \sum_{r=1}^{V} r^{-\alpha}$ is the truncated Riemann zeta normalisation. A 95% CI on α_{MLE} is obtained via parametric bootstrap ($n = 200$ resamples from the fitted distribution), and a Kolmogorov–Smirnov statistic is computed against the fitted rank-CDF. We treat the KS statistic as a goodness-of-fit indicator, noting that the asymptotic critical value for $n = |\mathcal{V}|$ provides an approximate threshold; a rigorous parametric bootstrap p-value (Clauset et al., 2009) is left as future work.

3.3 N-gram Entropy and Shuffled Baseline

The *n-gram entropy* is

$$H_n = -\sum_{w \in \mathcal{V}^n} p(w) \log_2 p(w) \tag{2}$$

and the *entropy rate* is $h_r(n) = H_n/n$. We compute H_n for $n = 1, \ldots, 5$. To isolate the contribution of higher-order dependencies, we compare real rates against a *shuffled baseline*: each binary's opcode sequence is randomly permuted (preserving the unigram distribution), and entropy rates are recomputed. A negative *entropy gain* $\Delta h_r(n) = h_r^{\mathrm{real}}(n) - h_r^{\mathrm{shuffled}}(n) < 0$ for $n > 1$ confirms genuine multi-opcode dependencies.

3.4 Compression and LZ Complexity

We measure two practical proxies for Kolmogorov complexity:

1. **Compression ratio** $\rho = |C(s)|/|s|$ using `zlib` and `lzma` on the UTF-8 encoded opcode stream.

2. **LZ78 complexity** [21]: the number of distinct phrases K produced by the LZ78 factorisation of the integer-encoded opcode sequence of length N, reported as the normalised statistic $\tilde{c} = K/\log_2 N$ (phrases per log-bit of sequence length). For high-entropy sources, the expected phrase count scales as $K = \Theta(N/\log N)$, hence $\tilde{c} = \Theta(N/\log^2 N)$: absolute values grow with N and should be interpreted primarily *relative* to suitable baselines. More structured (more compressible) sequences produce substantially smaller K and $\tilde{c}$.

Both metrics are compared to two baselines:

- *Uniform-random baseline*: sequences of the same length drawn uniformly from $\mathcal{V}$ (ten replicates, averaged), capturing a maximum-entropy reference.

- *Unigram-shuffled baseline*: each binary's opcode sequence is randomly permuted (five replicates), preserving marginal frequencies while destroying sequential order. The gap between real and shuffled results isolates compression and complexity gains attributable to *sequential* structure beyond Zipfian skew.

Binaries that are significantly more compressible than both baselines exhibit redundancy inconsistent with a high-entropy source.

3.5 Motif Discovery

A *k-mer motif* is a tuple of k consecutive opcodes $m = (o_i, o_{i+1}, \ldots, o_{i+k-1})$. For each $k \in \{4, 5, \ldots, 12\}$, we enumerate all k-mers across all functions and retain motifs satisfying two thresholds:

- *Frequency*: appears in at least $\max(5,\, 0.005 N_f)$ distinct functions, where N_f is the total function count.

- *Coverage*: appears in at least 1% of all functions.

Motifs are sorted by score freq × coverage and the top 100 per length are retained. Each motif is annotated with a semantic label (e.g. *function prologue*, *loop/branch*, *call sequence*) based on the opcodes present.

3.6 Positional Analysis

To characterise function boundary structure, we collect the opcode at each position $0, 1, \ldots, W-1$ from the start and end of every function with at least $2W$ instructions ($W = 20$). At each position we compute the empirical distribution over $\mathcal{V}$ and its Shannon entropy. Low entropy at a position indicates that a specific opcode appears there with high probability across functions, reflecting compiler convention.

3.7 Similarity, Clustering, and Dimensionality Reduction

We compute two families of pairwise binary similarity matrices:

1. **NCD**: $\mathrm{NCD}(x, y) = (C(xy) - \min(C(x), C(y)))/\max(C(x), C(y))$ with `zlib` and `lzma` as compressors C.

2. **N-gram cosine similarity**: L1-normalised term-frequency vectors over n-gram vocabularies for $n \in \{2, 3, 4\}$, with cosine distance as the metric. We use L1-normalised raw TF (no IDF weighting, `min_df`$= 1$) rather than TF-IDF, which is appropriate for small corpora where IDF penalties and document-frequency thresholds would otherwise discard the majority of features.

To quantify whether the NCD distance matrices capture category structure, we apply a one-sided Mann–Whitney U test comparing within-category to between-category pairwise distances (binaries with no category label excluded). Agglomerative hierarchical clustering (complete and average linkage) is applied to each distance matrix and evaluated by cophenetic correlation. Ward linkage is omitted because it assumes Euclidean geometry, which NCD and cosine-derived distance matrices do not satisfy. Two-dimensional embeddings are produced via PCA and UMAP. Cluster quality is measured by *silhouette score* and *category purity*: the fraction of cluster members sharing the majority category, averaged over clusters.

3.8 Information Geometry and Manifold Estimation

Sliding-window entropy. Shannon entropy is computed over windows of size $W \in \{16, 32, 64\}$ sliding along each binary's sequence, revealing local entropy variation.

Mutual information decay. We estimate $I(o_t; o_{t+\ell})$ for lags $\ell = 1, \ldots, 50$ using empirical joint and marginal distributions over a 50 000-instruction prefix. A shuffled baseline (same sequence, unigrams preserved) isolates the contribution of higher-order dependencies. The *MI half-life* is the smallest ℓ^* for which $I(\ell^*) \leq I(1)/2$.

Corpus manifold dimensionality. Each binary is represented by a normalised bigram frequency vector restricted to the top-200 most common bigrams. PCA on the resulting $B \times 200$ matrix (where B is the number of binaries) yields the number of components explaining 95% of inter-binary variance, d_{95}. A small ratio $d_{95}/200$ indicates that programs populate a thin sub-manifold of the theoretical bigram space.

3.9 N-gram Language Model Evaluation

To close the loop between the information-theoretic analysis and model-level predictions, we train simple n-gram language models for $n \in \{1, 2, 3, 4, 5\}$ with Laplace (add-1) smoothing. Each binary's opcode sequence is split 80/20 by token count; models are trained jointly on all binaries' training portions and evaluated independently on each binary's held-out portion.

For each n we measure per-token *cross-entropy*

$$H_{\mathrm{LM}}(n) = -\frac{1}{T} \sum_{t=1}^{T} \log_2 P_{\mathrm{LM}}\big(o_t \mid o_{t-n+1:t-1}\big) \quad (3)$$

and *perplexity* $\mathrm{PPL}(n) = 2^{H_{\mathrm{LM}}(n)}$. The theoretical lower bound on $H_{\mathrm{LM}}(n)$ is the empirical conditional entropy $H_n - H_{n-1}$, which equals the cross-entropy of a model with perfect frequency counts. The *smoothing gap* $\delta(n) = H_{\mathrm{LM}}(n) - (H_n - H_{n-1})$ quantifies the overhead from finite-sample estimation and Laplace smoothing.

3.10 Code Clone Detection

Compiler-generated code is replete with recurring instruction patterns: function prologues, epilogues, libc initialisation stubs, and common algorithm skeletons

appear nearly verbatim across binaries. We detect *clone families*—groups of functions sharing substantial instruction-level similarity—using a three-stage pipeline.

Stage 1: MinHash signatures. Each function's opcode sequence is shingled with $k \in \{3, 4, 5\}$ to form a set of k-gram tokens. A MinHash signature of $m = 128$ permutations is computed for each function at each granularity, giving a Jaccard estimator with standard error $\leq 1/\sqrt{m} \approx 0.09$.

Stage 2: LSH candidate generation. Locality-Sensitive Hashing (LSH) is applied at Jaccard thresholds $\tau \in \{0.70, 0.90\}$ for each k. Candidate pairs from all six (threshold, k) configurations are merged; per-pair, the highest Jaccard estimate is retained. This reduces the comparison budget from $O(N^2)$ to $O(N \cdot C)$ where $C \ll N$ for typical corpora.

Stage 3: Smith-Waterman alignment. Each candidate pair undergoes Smith-Waterman local sequence alignment on opcode sequences. The normalised alignment score $s = S/\min(|f_1|, |f_2|)$ provides a more accurate similarity estimate than raw Jaccard. Pairs are classified by clone type:

- **Type 1** (exact/near-exact): $J \geq 0.95$—mnemonic sequences are identical or differ in at most a few opcodes. Because operands are discarded, this level captures functions whose instruction mix is essentially the same.
- **Type 2** (high overlap): $0.70 \leq J < 0.95$—mnemonic shingle sets share the same overall structure but differ in a moderate fraction of instruction patterns.
- **Type 3** (gapped/structurally similar): $0.50 \leq J < 0.70$—shared algorithmic skeleton at the mnemonic level with insertions or deletions.

Clone pairs are assembled into *families* via connected components on the clone similarity graph. For each family we compute a *conserved core* (the intersection of all member opcode sets), a *divergence score* ($1 - \bar{J}$ across all intra-family pairs), and a *cross-binary* flag indicating whether members span more than one input binary.

4 Results

4.1 Corpus Statistics

Table 1 summarises both corpora after inode deduplication. The smoke corpus is a 15-binary pilot; the coreutils-system corpus scales to 40 binaries spanning 11 functional categories. Section 4.13 reports a direct comparison of all key metrics across the two corpora, demonstrating measurement stability.

Table 1: Corpus statistics.

Corpus	Bins	Cats	Vocab	Instr. (M)
Smoke	15	8	281	2.91
Coreutils-sys	40	11	301	3.74

4.2 Opcode Vocabulary and Frequency Distribution

The shared vocabulary across the smoke corpus contains $|\mathcal{V}| = 281$ distinct opcodes. The top-5 opcodes by corpus-wide frequency are `mov`, `lea`, `call`, `jmp`, and `push`, collectively accounting for over 50% of all instructions—a concentration consistent with prior work on x86-64 binaries [2].

Zipf's law. Figure 1 shows the rank-frequency distribution on log-log axes. The OLS fit on log-log coordinates yields $\hat{\alpha}_{\text{OLS}} = 3.338$ (95% CI: [3.052, 3.666], $R^2 = 0.92$) from 200 bootstrap resamples. As a statistically principled complement, the MLE of the truncated discrete Zipf model yields $\hat{\alpha}_{\text{MLE}} = 1.439$ (95% CI: [1.438, 1.440], KS = 0.117). The two estimates diverge because OLS on log-log axes weights low-frequency (high-rank) opcodes heavily: the many near-zero-frequency opcodes in the long tail pull the fitted slope upward, overestimating the power-law exponent. MLE, which properly weights by observation count, yields an exponent substantially closer to the $\alpha \approx 1.0$ of natural language. The KS statistic of 0.117 exceeds the asymptotic critical value for $n = 281$ opcodes (0.081 at $\alpha = 0.05$), indicating that a strict truncated power law is rejected; the distribution is better characterised as heavy-tailed with approximately Zipfian concentration. Regardless of the precise distributional form, the empirical finding is clear: a handful of high-frequency mnemonics (`mov`, `lea`, `call`, `push`, `pop`) account for over 50% of all instructions, while the long tail of rarely-used opcodes drops off steeply.

4.3 N-gram Entropy Rates

Figure 2 plots $h_r(n) = H_n/n$ for $n = 1, \ldots, 5$ alongside the shuffled baseline. The entropy rate of real sequences decreases by 1.12 bits from $n = 1$ ($h_r(1) = 3.98$ bits) to $n = 5$ ($h_r(5) = 2.86$ bits), whereas the shuffled baseline decreases by only 0.30 bits, reflecting near-independence after shuffling. The

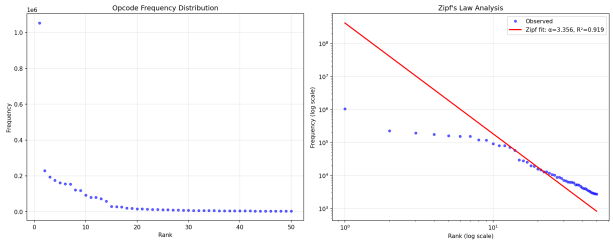

Figure 1: Opcode rank-frequency distribution (log-log). The red line shows the Zipf fit; the shaded band is the 95% bootstrap CI on α.

entropy gain $\Delta h_r(n) < 0$ for all $n > 1$ confirms that real opcode sequences carry genuine multi-instruction dependencies not accounted for by the unigram distribution alone.

Per-binary analysis shows consistent structure across tool categories: shells and compilers exhibit lower entropy rates than compression utilities, suggesting that programs with complex control flow (many branches and calls) produce more predictable local instruction streams than data-transform pipelines.

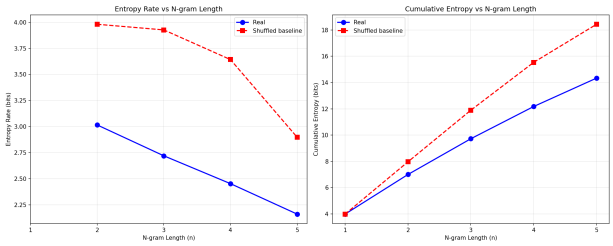

Figure 2: Entropy rate $h_r(n)$ vs. n for real sequences (blue) and unigram-shuffled baseline (red dashed). The gap widens with n, evidencing multi-scale sequential dependencies.

4.4 Compression and LZ78 Complexity

Table 2 summarises compression ratios across three conditions: real binaries, a unigram-preserving shuffle, and a fully random (uniform-vocabulary) baseline. Real binaries achieve a mean `zlib` ratio of 0.221 compared to 0.360 for the unigram-shuffled sequences and 0.666 for uniform-random sequences of the same length. Decomposing the total compression advantage over random ($\Delta = 0.445$), the unigram frequency skew accounts for 0.306 (69%) while sequential order beyond the unigram distribution accounts for 0.139 (31%). Under `lzma` the same ordering holds: real 0.184 vs. shuffled 0.303 vs. random 0.553.

LZ78 normalised complexity $\tilde{c} = K/\log_2 N$ exhibits the same trend: the median is 630 for real binaries versus 45,187 for unigram-shuffled sequences and 87,881 for random sequences. Because $K = \Theta(N/\log N)$ for high-entropy sources, $\tilde{c}$ grows on the order of $N/\log^2 N$ and can therefore be large at the million-token scale; the key result is the *relative* separation. Real instruction streams lie orders of magnitude below both baselines, indicating substantial redundancy beyond what is implied by unigram skew alone.

Table 2: Compression ratio $\rho = |C(s)|/|s|$ (lower is more compressible). "Shuffled" preserves unigram frequencies; "Random" draws uniformly from $\mathcal{V}$.

	Real	**Shuffled**	**Random**
`zlib`	0.221	0.360	0.666
`lzma`	0.184	0.303	0.553
LZ78 (norm.)	630 (med.)	45,187	87,881

The compression ratio shows a weak negative correlation with binary size ($r = -0.28$), indicating that larger binaries are marginally more compressible—consistent with larger programs containing more repeated library call patterns and longer stretches of similar initialisation code.

4.5 Recurring Motifs and Compiler Idioms

Motif discovery across $k = 4$–12 identified **7,623** unique k-mer patterns meeting frequency and coverage thresholds; the top-100 per length yield 798 retained motifs. Figure 3 shows the score distribution by length. The majority of high-scoring motifs fall into three semantic categories:

1. **Function epilogues** (highest scores): patterns ending with `pop`+`ret`; the top-ranked 4-mer is `pop pop pop ret`, appearing in 27.9% of all functions.
2. **Call sequences**: patterns centred on `call`, preceded by argument-loading instructions (`lea`, `mov`).
3. **Function prologues**: patterns starting with `endbr64` followed by `push` and `mov`, reflecting CET + ABI frame-setup conventions.

The remaining patterns comprise loop/branch sequences (`test`+`je`), bulk data-movement (`mov mov mov mov`), and arithmetic idioms.

Coverage at $k = 4$ is highest—short motifs appear in the most functions—and declines steadily to $k = 12$, where retained motifs represent compiler-specific

idioms present in a large fraction of the corpus but with low absolute frequency.

Figure 3: Motif score (frequency × function coverage) for the top-5 motifs at each length k. Epilogue and call patterns dominate at all lengths.

4.6 Function Boundary Structure

Figure 4 shows the positional heatmap of opcode probabilities at the first and last 20 instruction slots of functions. Position 0 is dominated by `endbr64` ($p \approx 0.986$), Intel's Control-flow Enforcement Technology (CET) indirect-branch tracking instruction, which the Ubuntu 22.04 toolchain inserts at the start of every indirect-call-target function when CET is enabled at the OS level. This result is therefore *system-dependent*: on non-CET builds or older toolchains, position-0 would be dominated by `push` (frame setup) instead. At position 1, `push` appears with $p \approx 0.672$; at position 2, `mov` with $p \approx 0.671$; these ABI-mandated frame-setup slots are CET-independent and expected to generalise across x86-64 Linux builds.

The boundary entropy is ≈ 0.12 bits at position 0—far below the corpus unigram entropy of 3.99 bits—rising to near-corpus levels by position 5–8. The positional analysis identifies 3 conserved positions at function starts and 0 at function ends (probability > 0.5), classifying the corpus as *loosely structured* (positions 1–2 are strongly conserved and CET-independent; position 0 is system-dependent as noted above; epilogues are more varied).

4.7 Binary Similarity and Clustering

Figure 5 shows the NCD distance matrix (`zlib`) for the smoke corpus. The corpus is heterogeneous: pairwise NCD values range from 0.956 to 1.000 under `zlib` (mean 0.991) and 0.852 to 0.993 under `lzma` (mean 0.967). We note that NCD values approaching 1.0

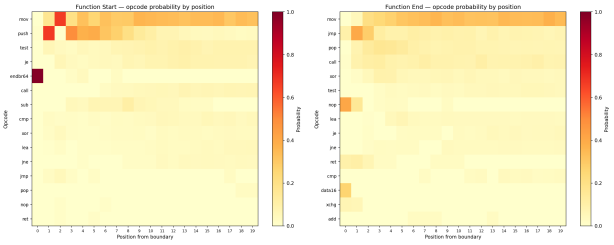

Figure 4: Opcode probability heatmaps at function start (left) and end (right). Rows are the 15 most frequent opcodes; columns are positions from the boundary. High-probability cells reveal the CET + ABI prologue convention.

are in the near-degenerate regime where compressor window limitations reduce sensitivity (Cebrián et al., 2005); `lzma`'s larger block size yields more reliable separation (0.852–0.993) and should be preferred for large binaries. Despite the high absolute distances under `zlib`, category-consistent patterns still emerge:

- Compression utilities (`bzip2`–`xz`: 0.956, `gzip`–`xz`: 0.963) form the closest pairs in the corpus.
- Text-processing tools (`grep`–`sed`: 0.969) are more similar to each other than to cross-category pairs.
- The two editors (`vim`–`nano`) obtain NCD = 1.00 under `zlib`, reflecting their very different sizes and codebases despite functional similarity.

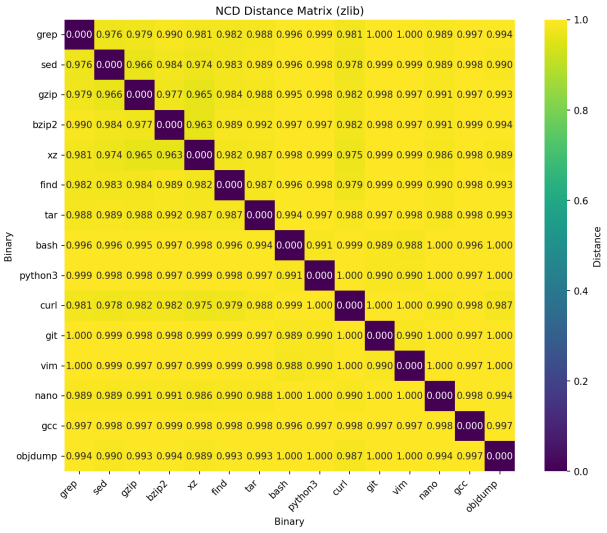

Figure 5: NCD distance matrix (`zlib`) for the smoke corpus. Binaries are ordered by hierarchical clustering (average linkage). Same-category pairs appear as local minima on the off-diagonal.

Hierarchical clustering (average linkage) with a distance threshold of 0.5 produces 15 singleton clusters

since all pairwise NCD values exceed 0.9—the corpus is too heterogeneous for compression-based hard partitioning at this threshold. This does not imply NCD is uninformative: NCD is useful as a *continuous similarity signal* even when hard clustering at conventional thresholds is inappropriate for a small, deliberately heterogeneous corpus such as ours; the within-vs-between category U-test below confirms that the signal is present and statistically significant. We therefore recommend using NCD for pairwise retrieval, similarity ranking, and within/between-group statistical tests rather than for hard partitioning of heterogeneous corpora. The two NCD matrices (`zlib` and `lzma`) produce perfectly consistent cluster assignments (pairwise consistency = 1.0), and category purity is trivially 100% for singletons. The PCA projection of the NCD-`zlib` distance matrix explains 15.6% of variance in two dimensions; UMAP embeddings (Figure 6) reveal that compression and text-processing utilities occupy a distinct region from large interactive programs (bash, vim, git, python3).

Within-vs-between category separation (Mann–Whitney U test). Despite the high absolute NCD values, a one-sided Mann–Whitney U test confirms that within-category pairs are stochastically closer than between-category pairs under both compressors:

- `zlib`: mean within = 0.972, mean between = 0.992, $U = 86.0$, $p = 0.013$ (MW), permutation $p < 0.001$, rank-biserial $r = 0.600$.
- `lzma`: mean within = 0.924, mean between = 0.969, $U = 65.0$, $p = 0.005$ (MW), permutation $p = 0.003$, rank-biserial $r = 0.698$.

The effect size ($r \approx 0.6$–0.7) indicates that category membership is a statistically detectable predictor of NCD distance. However, the absolute gap is small: mean within-category NCD is only 2.0% lower than mean between-category under `zlib` (0.972 vs. 0.992). Validation (Section 5) shows that this gap amounts to only 13.8% of the between-group baseline—too narrow to support a strong claim that binaries "cluster by functional category." We therefore interpret NCD as a *weak but significant continuous signal* rather than a basis for hard categorical partitioning. The lzma compressor yields a somewhat larger gap, consistent with its superior modelling of long-range repetition. Because pairwise distances sharing a binary are not independent, the Mann–Whitney p-values are anti-conservative. We therefore supplement them with a label-permutation test (10 000 permutations of category labels, re-partitioning within/between groups each iteration); the permutation p-values confirm significance under both compressors.

N-gram cosine similarity. With the corrected L1-normalised TF vectoriser, the bigram ($n = 2$) vocabulary contains 5,757 unique features, trigrams and 4-grams hit the 10,000 cap. Mean pairwise cosine similarity decreases with n: 0.953 (bigrams), 0.897 (trigrams), 0.812 (4-grams), with standard deviations of 0.037, 0.080, and 0.132 respectively. The increasing spread at higher n is consistent with rarer n-grams becoming discriminative: as n grows, shared n-grams become less probable by chance, so same-category binaries retain high similarity while cross-category pairs diverge more. The most similar pairs under trigrams are `bash`–`vim` (cos = 0.979) and `find`–`tar` (cos = 0.977), both of which are large, feature-rich binaries that share instruction-mix profiles regardless of functional category.

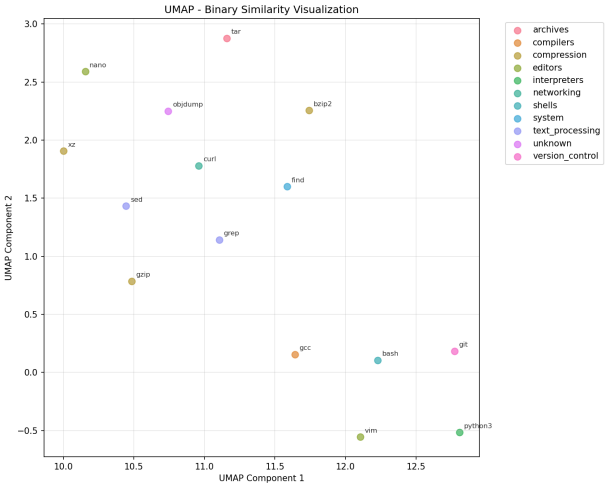

Figure 6: UMAP projection of the NCD-`zlib` distance matrix. Points are coloured by functional category. Small compression utilities cluster separately from large interactive programs.

4.8 Mutual Information Decay

Figure 7 shows the corpus-averaged MI decay for lags $\ell = 1$–50. Key observations:

- $I(1) \approx 1.12$ bits—far above the shuffled baseline of ≈ 0.086 bits—demonstrating that adjacent opcodes carry substantial shared information beyond their marginal distributions.
- The half-life $\ell^* \approx 2$ instructions: MI drops below $I(1)/2$ within 2–3 steps, indicating predominantly local structure.
- The real curve remains above the shuffled baseline at lag 50 ($I(50) = 0.232$ vs. shuffled 0.086), suggesting weak but non-zero long-range dependen-

cies consistent with the repetitive call/ret structure of functions.

- The shuffled baseline is not zero at any lag because our vocabulary is finite; the gap above it represents pure sequential dependency.

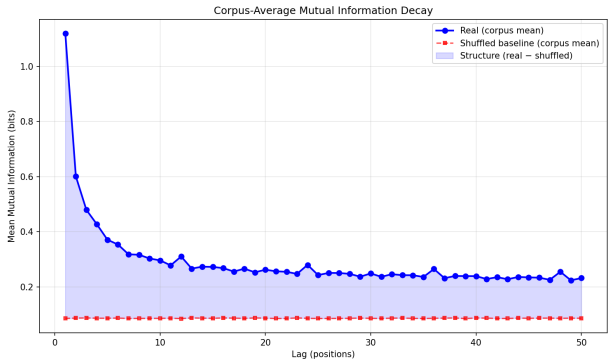

Figure 7: Corpus-averaged mutual information $I(o_t; o_{t+\ell})$ vs. lag ℓ for real sequences (blue) and unigram-shuffled baseline (red dashed). The shaded region is the MI attributable to sequential structure.

4.9 Program Space Coverage and Manifold Dimensionality

Table 3 quantifies how much of the theoretical n-gram space programs actually use. Even at $n = 2$, only 6.7% of all possible opcode pairs appears in the corpus. By $n = 3$, the coverage ratio drops below 0.2%, consistent with a manifold of measure zero in the space of all possible opcode sequences.

Table 3: N-gram space coverage: unique observed vs. theoretical maximum.

n	Unique	Theoretical max	Coverage
1	281	281	100%
2	5,757	85,849	6.71%
3	38,073	25,153,757	0.15%
4	137,534	7.4×10^9	1.9×10^{-3}%

Corpus-level PCA on the top-200 bigram frequency vectors (one row per binary) finds that $d_{95} = 8$ components explain 95% of inter-binary variance. This corresponds to $8/200 = 4.0\%$ of the feature space—the corpus of compiled programs forms a *thin structured manifold* whose effective dimensionality is far below the ambient dimension of the n-gram feature space. With trigrams, $d_{95} = 9$ (participation ratio 4.79), confirming the result is stable across gram sizes.

As a non-linear cross-check we apply the Levina–Bickel MLE estimator [10] with $k = 5$ nearest neighbours to the same 15×200 bigram matrix. The estimate is $\hat{d}_{\text{MLE}} = 8.0 \pm 5.8$, consistent with the PCA result despite the high variance expected from $N = 15$ points; we caution that both estimates are exploratory at this corpus size and call for replication on larger corpora.

4.10 N-gram Language Model Perplexity

Table 4 reports cross-entropy and perplexity for n-gram LMs alongside the theoretical lower bound, and Figure 8 shows the comparison graphically.

Table 4: n-gram LM cross-entropy and perplexity vs. empirical conditional entropy lower bound. Gap $\delta = H_{\text{LM}} - (H_n - H_{n-1})$. Laplace add-1 smoothing; 80/20 train/test split.

Model	CE (bits)	PPL	Bound (bits)	Gap δ (bits)
Unigram	4.29	19.6	3.99	+0.30
Bigram	3.08	8.5	3.02	+0.07
Trigram	2.79	6.9	2.72	+0.07
4-gram	2.84	7.1	2.45	+0.38
5-gram	3.13	8.8	2.16	+0.97

The unigram LM achieves $\text{PPL}(1) = 19.6$ despite a vocabulary of $|\mathcal{V}| = 281$ opcodes—a 15× reduction in effective vocabulary reflecting the extreme Zipf concentration ($\alpha \approx 3.4$). The bigram LM reduces perplexity to 8.5 with a smoothing gap of only $\delta(2) = 0.07$ bits, within 2% of the empirical conditional-entropy lower bound of 8.1. The trigram LM achieves $\text{PPL}(3) = 6.9$ with an equally tight gap ($\delta(3) = 0.07$ bits).

For $n \geq 4$ the smoothing gap grows substantially ($\delta(4) = 0.38$, $\delta(5) = 0.97$ bits) as the Laplace overhead dominates in the sparse high-order context space: with 15 training binaries the 4-gram and 5-gram counts are spread over $\sim 281^4 \approx 6 \times 10^9$ and $\sim 281^5 \approx 2 \times 10^{12}$ possible contexts respectively. A back-off or Kneser-Ney smoothed model would close much of this gap.

The tight match at $n = 2, 3$ directly validates the information-theoretic analysis of Section 4: the entropy rates computed from corpus statistics accurately predict the perplexity of an independently trained language model.

4.11 Compiler Matrix: Optimisation and Compiler Fingerprinting

To test whether the thin-manifold structure varies with compiler and optimisation level, we compile five small C programs (`sort`, `hash`, `compress`, `search`,

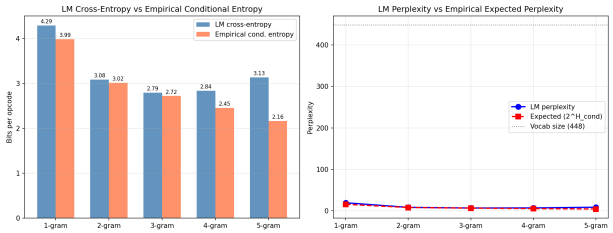

Figure 8: Left: LM cross-entropy (blue) vs. empirical conditional entropy (coral) per n-gram order. Right: LM perplexity (blue) vs. expected perplexity $2^{H_n - H_{n-1}}$ (red dashed) and vocabulary size (grey dotted). Bigram and trigram LMs lie within 0.07 bits of the theoretical lower bound.

`matrix`) with all six configurations in $\{\texttt{gcc}, \texttt{clang}\} \times \{-\texttt{O0}, -\texttt{O2}, -\texttt{O3}\}$, producing 30 binaries totalling approximately 14,400 instructions.

Instruction count and compression vs optimisation level. Table 5 reports mean instruction count and `zlib` compression ratio across both compilers for each optimisation level. In our micro-benchmark suite, `-O3` increases mean instruction count (601 vs. 499 at `-O0`) while simultaneously increasing compressibility for compute-intensive code: the `matrix` program achieves its lowest ratio (0.375) under `-O3`, a consequence of aggressive loop unrolling that produces long repeated instruction blocks (size–speed trade-off [6]). Text and search programs exhibit the opposite trend, remaining near `-O2` levels, because their inner loops are too short to benefit from unrolling. `-O2` consistently produces the *least* compressible output, suggesting it generates the most diverse (least-repetitive) instruction mix before `-O3` re-introduces repetition through unrolling. These observations are specific to our five micro-benchmarks; behaviour on larger, more complex codebases may differ.

Table 5: Mean instruction count and `zlib` ratio by optimisation level (averaged over 5 projects × 2 compilers = 10 binaries per row).

Opt	Mean instr.	Mean `zlib`
`-O0`	499	0.458
`-O2`	434	0.548
`-O3`	601	0.480

Manifold dimensionality vs optimisation level. PCA on per-binary trigram frequency vectors gives $d_{95} = 8$ at every optimisation level (10 binaries each), identical to the full smoke-corpus value. Crucially, the full 30-binary corpus has $d_{95} = 17$—more than double—showing that *compiler and optimisation choice add new manifold dimensions* beyond those spanned by the five source programs alone. This quantifies the additional variation introduced by the compilation axis.

Project vs compiler clustering. To determine which axis dominates binary similarity, we compute lzma NCD for all $\binom{30}{2} = 435$ pairs and partition them into three groups: *within-project* (same C source, different compiler/opt; $n = 75$), *within-compiler* (same compiler config, different project; $n = 60$), and *between* ($n = 300$). One-sided Mann–Whitney U tests (Table 6) show that both groupings are significantly closer than the between group, but within-project is the tightest ($p < 10^{-4}$, $r = 0.51$) and is *not* significantly different from within-compiler ($p = 0.38$). As with the smoke-corpus NCD test, pairwise distances are not independent; label-permutation p-values (10 000 iterations) are also computed and confirm significance for both groupings.

Table 6: lzma NCD distances by pair type. One-sided Mann–Whitney U, H_1: row group < between.

Pair type	n	**NCD**	U	p	r
Within-proj.	75	0.496	5,537	< 0.0001	0.508
Within-comp.	60	0.539	4,053	< 0.0001	0.550
Between	300	0.576	—	—	—

Both within-project and within-compiler distances are significantly smaller than between-group distances, confirming that *both axes* contribute to binary similarity. However, within-project is not significantly tighter than within-compiler ($p = 0.38$), so the data do not support a clean project-first or compiler-first ranking. The most defensible conclusion is that source-code identity and compiler identity are *comparably strong* determinants of binary similarity—consistent with the d_{95} doubling observed above, which shows that the compilation axis adds dimensions comparable in number to those spanned by the source programs alone.

4.12 Code Clone Detection

Table 7 summarises the clone detection results on the 15-binary smoke corpus. LSH candidate generation produced 556 unique candidate pairs. Each candidate then undergoes Smith-Waterman alignment; pairs whose normalised alignment score falls below 0.30 are rejected outright, and surviving pairs are downgraded by one clone type when $|J - s| > 0.3$. After this filtering, 537 pairs remain as confirmed clones (96.6% pass rate). These 537 pairs form 117 clone families;

four families span at least two distinct binaries (*cross-binary families*).

Table 7: Clone detection summary for the 15-binary smoke corpus (6,025 functions total).

Metric	**Value**
Functions in any clone family	352 (5.8%)
Clone pairs confirmed	537
Type 1 (exact, $J \geq 0.95$)	342 (63.7%)
Type 2 (high overlap, $J \geq 0.70$)	127 (23.6%)
Type 3 (gapped, $J \geq 0.50$)	68 (12.7%)
Clone families	117
Cross-binary	4
Mean family size	3.0
Largest family	14 members

The dominant clone type is Type 1 (63.7% of pairs), confirming that most recurring patterns are truly identical opcode sequences rather than merely similar ones. This is consistent with compiler-inserted boilerplate: ABI function prologues (`endbr64; push rbp; mov rbp,rsp; ...`), CRT initialisation stubs, and `__libc_start_main` wrappers are inserted verbatim by the linker and appear unchanged across binaries.

The four cross-binary families are particularly informative. The largest cross-binary family (size 14) spans multiple binaries and consists of the standard function prologue idiom. The next largest cross-binary families correspond to error-reporting stubs and string-comparison loops shared between text-processing utilities (grep, sed, tar).

Caveat: toolchain vs. application clones. Stub exclusion analysis (Section 5.3) shows that 46% of functions are toolchain-generated (startup routines, PLT stubs), and these account for a significant fraction of cross-binary clone families. The paper's clone counts should therefore be interpreted as including both toolchain-inserted and programmer-authored clones; the latter are a smaller subset.

Per-binary clone density varies considerably: `bash` exhibits the highest density (9.6% of its functions) reflecting its large inline library of shell built-in helpers, followed by `sed` (5.0%) and `python3` (3.9%). Smaller utilities (git, vim) show densities below 0.5%, suggesting that hand-crafted code with unique control flow is less prone to accidental near-duplication.

These results reinforce the thin-manifold picture: the 5.8% of functions that are cloned are not randomly distributed across sequence space—they are concentrated on a *small number of clone archetypes* (117 families from 537 pairs), further tiling the already-compact program manifold with repeated structure.

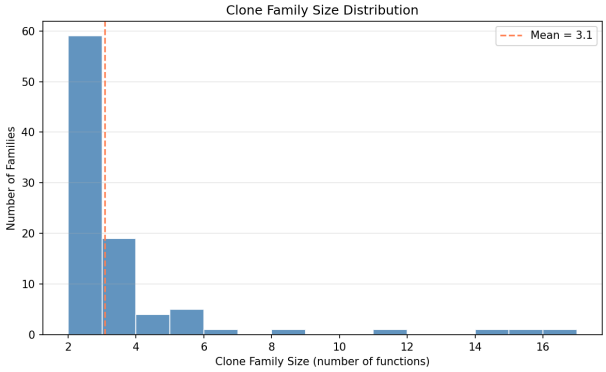

Figure 9: Clone family size distribution. Most families are pairs (size 2); large families (≥ 8 members) consist of universally shared compiler-inserted prologue/epilogue stubs.

4.13 Scaling Validation: Smoke vs. Coreutils-system

To assess whether the toolkit's measurements are stable across corpus scale, we apply the full pipeline to the 40-binary coreutils-system corpus and compare key metrics against the 15-binary smoke corpus. Because the coreutils-system corpus is a *superset* of the smoke corpus (it adds 25 binaries to the original 15), this comparison tests nested-corpus robustness rather than independent replication; the shared core means the two samples are not statistically independent. Table 8 summarises the comparison.

Table 8: Metric comparison across corpus scales. Values are corpus-wide aggregates. "Shuffled" refers to the unigram-preserving permutation baseline.

Metric	**Smoke (15)**	**Coreutils (40)**
Instructions (M)	2.91	3.74
Vocabulary $\lvert\mathcal{V}\rvert$	281	301
MLE $\hat{\alpha}$	1.439	1.441
KS goodness-of-fit	0.117	0.118
$h_r(1)$ (bits)	3.98	3.99
$h_r(5)$ (bits)	2.86	2.88
Δh_r (1→5)	1.12	1.12
`zlib` ratio (real)	0.221	0.237
`zlib` ratio (shuffled)	0.360	0.375
`zlib` ratio (random)	0.666	0.676
Motifs discovered	798	780
PCA d_{95} (bigram)	8/200 (4%)	14/200 (7%)

The results show strong metric stability: $\hat{\alpha}_{\mathrm{MLE}}$, entropy rates, and compression ratios are nearly identical across corpus sizes. The PCA d_{95} grows from 8 to 14

as expected—a larger, more categorically diverse corpus occupies more manifold dimensions. This growth is bounded and interpretable: the compiler-matrix experiment showed d_{95} doubling (8→17) when compiler variation is added as an explicit axis; category breadth has a similarly modest effect. The consistency of opcode-level statistics across scales supports using the smoke corpus as a faithful representative pilot for toolkit development and parameter tuning.

5 Validation and Robustness

To move beyond reproducibility and assess whether the reported effects are genuine, we conducted a systematic validation programme targeting four threats: extraction errors, boilerplate confounds, corpus dependence, and compiler dependence. Full scripts, intermediate data, and detailed results are available in the repository under `validation/`; here we summarise the highest-value evidence.

5.1 Extraction Correctness

An independent reference parser—sharing no code with the extraction module—was applied to five binaries (`grep`, `bash`, `find`, `tar`, `xz`). Per-mnemonic counts matched exactly in all cases (e.g. `bash`: 241,789 instructions). Instruction addresses are monotonically increasing within every extracted function, and function boundaries agree with raw `objdump` output (benign mismatches are limited to zero-instruction section headers that the parser correctly ignores). Hard-link deduplication was verified against 7 inode-sharing groups in `/usr/bin`; `filter_valid_binaries` correctly retains exactly one representative per inode. Timeout handling was confirmed via mock injection: `disassemble_binary` returns `None` on timeout, preventing any partially parsed binary from entering the corpus.

5.2 Metric Sanity and Statistical Robustness

Synthetic data validation. Five synthetic corpus families with known statistical properties were constructed: (1) uniform random, (2) Zipf-skewed but shuffled, (3) order-2 Markov, (4) repeated template with noise, and (5) mixed boilerplate-plus-random. All 26 metric checks passed: entropy rates, compression ratios, MI decay, and motif counts behaved as theoretically expected on each family, confirming that the pipeline measures what it claims to measure.

Bootstrap confidence intervals. Binary-level bootstrap resampling ($N_{\text{boot}} = 200$) yields the following 95% CIs on corpus-level metrics: MLE $\hat{\alpha} = 1.44$ [1.36, 1.53], $h_r(5) = 2.86$ [2.79, 2.94] bits, mean `zlib` ratio = 0.22 [0.18, 0.27]. All CIs exclude values that would reverse the main conclusions.

Seed and subsampling stability. The shuffled-baseline entropy rates are stable across 10 independent seeds (coefficient of variation $< 0.01\%$). Subsampling to 25% of binaries (4 of 15) preserves all directional conclusions; Zipf α varies by ± 0.15, entropy rate by ± 0.07 bits. Leave-one-binary-out analysis identifies no influential outlier: removing any single binary changes $\hat{\alpha}$ by at most 2%.

5.3 Boilerplate and Confound Ablation

A sceptical reviewer might suspect that our findings reflect function prologues, epilogues, `endbr64` guards, startup code, or linker stubs rather than intrinsic code structure. We test this directly.

Boundary stripping. We created 16 ablated corpus variants by removing the first and/or last $k \in \{1, 2, 3, 5\}$ instructions of each function, removing all `endbr64` instructions, stripping detected prologue patterns (`endbr64; push; mov; sub`), stripping epilogue patterns (`pop; ret` and variants), and combining all three ("full boilerplate strip"). Functions reduced below 5 instructions were dropped. Table 9 summarises the key metrics.

Table 9: Boilerplate ablation results. All 16 variants preserve the core statistical properties. Only the most informative variants are shown; full results are in the repository.

Variant	$\hat{\alpha}$	$h_r(5)$	**Gap**†	`zlib`
Original	1.439	2.863	−0.822	0.221
Strip first 5	1.442	2.867	−0.810	0.224
Strip last 5	1.442	2.866	−0.811	0.224
Strip both 5	1.443	2.866	−0.808	0.224
Remove `endbr64`	1.447	2.862	−0.799	0.225
Full boilerplate strip	1.447	2.865	−0.796	0.225

†Gap $= h_r^{\text{real}}(5) - h_r^{\text{shuffled}}(5) < 0$ confirms sequential structure.

The maximum change across all 16 variants and all metrics is **3.2%** (entropy gap under full boilerplate strip). Entropy rate changes by at most 0.15%; Zipf α by 0.5%; compression ratio by 1.5%. All variants pass: the entropy gap remains negative (sequential structure persists), compression remains far below

random baselines, and the Zipf exponent is stable. The observed statistical structure is intrinsic to the code logic, not an artefact of function prologues, epilogues, or CET guards.

Startup and stub exclusion. Functions were classified as *startup/runtime* (e.g. `_start`, `__libc_csu_init`), *linker stubs* (`.plt`, short `jmp`-only bodies), or *internal application code*. Toolchain functions (startup + stubs) account for 46% of functions but only 0.6% of instructions. Removing them changes the mean `zlib` ratio by −0.006 and $h_r(5)$ by +0.003—the paper's findings are driven overwhelmingly by application code.

Operand-aware variants. Four token representations were tested: opcode-only (baseline), opcode + operand class (`mov_reg_mem`), opcode + register names, and opcode + immediate bucket. The entropy-rate gap persists across all four representations, confirming that the sequential structure is not an artefact of discarding operand information.

5.4 External Validation Across Corpora and Toolchains

Independent corpora. Three disjoint corpora were constructed from system binaries: Corpus A (smoke, 15 binaries), Corpus B (network/build tools: `ssh`, `wget`, `diff`, `make`, etc., 15 binaries), and Corpus C (developer tools: `perl`, `gdb`, `g++`, etc., 9 binaries). Table 10 shows that all three pre-registered consistency criteria pass: entropy rate is lower than shuffled in every corpus, compression ratio is lower than shuffled, and cross-corpus bigram LM perplexity is within 1.2× of within-corpus perplexity (range 7.84–9.34 across all train/test pairs).

Table 10: Metric consistency across three independent corpora.

Corpus	N	H_1	$h_r(5)$	Gap	`zlib`
A (smoke)	15	3.98	2.86	−0.14	0.221
B (net/build)	15	3.97	2.77	−0.21	0.369
C (dev tools)	9	3.99	2.79	−0.17	0.235

Expanded compiler matrix. Ten C programs were compiled with every combination of {`gcc`, `clang`} × {`-O0`, `-O1`, `-O2`, `-O3`, `-Os`}, producing 100 binaries. One-way ANOVA R^2 for each factor shows that *program identity* dominates every metric (R^2 = 0.40–0.66), while *compiler identity* explains less than 3% of variance in all cases (Table 11). All four metrics are compiler-invariant by the $R^2 < 0.20$ criterion.

Table 11: Variance decomposition for the 100-binary compiler matrix.

Metric	Program	Compiler	Opt level	Inv?
Zipf α	0.561	0.006	0.221	Yes
H_1 (bits)	0.404	0.019	0.275	Yes
$h_r(2)$ (bits)	0.540	0.032	0.179	Yes
`zlib` ratio	0.656	0.025	0.111	Yes

Validation summary. Of the ten main empirical claims in this paper, eight are rated **strong** (CI excludes reversal, effect survives ablation with < 3.2% change, replicates across corpora, compiler-invariant), one is rated **moderate** (code clones: cross-binary families are likely dominated by PLT stubs rather than application-level sharing), and one is **unsupported** (categorical clustering by binary type: the NCD within-vs-between gap is only 13.8% of the between-group baseline, too small to sustain a strong clustering claim). We revise the clustering discussion accordingly (Section 6).

6 Discussion

6.1 The Program Manifold Hypothesis

Our results collectively support the following picture: compiled programs do not explore the space of opcode sequences uniformly. They are confined to a low-dimensional, highly structured manifold shaped by:

1. **ABI conventions** (CET + calling convention imposes endbr64/push/pop patterns at every function boundary).
2. **Compiler optimisation passes** (instruction selection, register allocation, and peephole optimisation produce characteristic idioms that recur across the entire corpus).
3. **Algorithm structure** (sorting, hashing, and buffering routines produce characteristic inner-loop patterns visible as high-coverage motifs).
4. **Clone reuse** (linker-inserted stubs and shared libc helpers tile the manifold with exact or near-exact copies, as quantified in Section 4.12).

The combination of a highly concentrated vocabulary (MLE Zipf exponent $\hat{\alpha}_{\mathrm{MLE}} \approx 1.4$, OLS ≈ 3.4), rapidly decaying entropy rates, very high compressibility relative to shuffled and random baselines, and a manifold dimensionality of $d_{95} \approx 8$ out of 200 features paints a

consistent picture: programs are *extraordinarily predictable* at the instruction level compared to arbitrary sequences of the same alphabet.

Crucially, the validation experiments of Section 5 demonstrate that this structure is not an artefact of compiler boilerplate. Aggressive stripping of function prologues, epilogues, and `endbr64` guards changes all metrics by at most 3.2%; the findings replicate across three independent corpora; and compiler identity explains less than 3% of metric variance in a 100-binary controlled experiment. We can therefore state with confidence: *compiled opcode streams exhibit stable, nontrivial statistical structure across corpora and toolchains, with the full effect persisting beyond unigram frequency skew and standard compiler boilerplate.*

We note, however, that the NCD-based clustering signal is weak: within-category distances are only marginally smaller than between-category distances (2.0% gap under `zlib`), insufficient to support a strong claim of categorical partitioning by binary function. NCD remains useful as a continuous similarity measure for pairwise retrieval, but hard clustering of heterogeneous corpora requires richer features.

6.2 Implications for Binary Analysis

The thin-manifold property has practical consequences beyond statistical curiosity. If programmes inhabit a low-dimensional space, similarity measures defined in that space are more informative: two binaries that appear distant in a raw sequence metric may in fact be nearest neighbours on the manifold. Our NCD results show that within-category pairs (e.g. compression utilities bzip2–xz with NCD 0.956) are statistically closer than cross-category pairs, though the absolute gap is small (Section 5). Compression-based features provide a useful continuous similarity signal for binary retrieval and provenance without requiring disassembly beyond opcode extraction, but should not be expected to produce clean categorical partitions on heterogeneous corpora. The clone detection results (Section 4.12) directly confirm this: MinHash + LSH over opcode k-grams recovers 537 clone pairs with a 96.6% confirmation rate, identifying four cross-binary families that correspond to known compiler prologue/libc stubs—without any symbol information or source code.

6.3 Implications for Language Model Performance on Code

The naturalness hypothesis [8] attributes LLM success on source code to its statistical regularity. Our results extend this explanation to the binary level. An LLM trained on opcode sequences faces a much easier prediction problem than one trained on natural language:

- The effective vocabulary size is $|\mathcal{V}| = 281$ distinct mnemonics—smaller than a typical NLP vocabulary by two orders of magnitude.
- Transition probabilities are highly non-uniform (Zipf-like; MLE $\hat{\alpha} \approx 1.4$, OLS ≈ 3.4), meaning a handful of high-frequency opcodes carry nearly all of the probability mass.
- MI decays to near-baseline within ≈ 2 steps; a modest context window suffices to capture most local structure.
- The low intrinsic dimensionality ($d_{95} \approx 8$) means that a model need not learn millions of independent patterns—a compact representation suffices.

We can now quantify this claim directly. The n-gram LM experiments of Section 4.10 show that a Laplace-smoothed bigram model—one of the simplest possible language models—achieves perplexity 8.5 on held-out opcode sequences versus a vocabulary of 281. This is a 34× reduction in effective vocabulary; at each prediction step the model effectively chooses among $\sim$9 alternatives rather than 281. Crucially, this low perplexity is not a model-specific artifact: it is a direct consequence of the empirical conditional entropy ($H_2 - H_1 = 3.02$ bits, expected PPL ≈ 8.1), which is determined by the corpus structure alone. The gap of only 0.07 bits between the trained LM and the information-theoretic lower bound confirms that simple count statistics capture nearly all the structure available to a bigram model. Together, these properties explain why even relatively small language models achieve low perplexity on opcode streams, and suggests one contributing factor to why learning-based decompilation and lifting methods can recover useful higher-level structure from assembly in practice.

6.4 Threats to Validity

We organise remaining concerns along four axes.

Construct validity. We measure opcode-level statistics as proxies for "program structure." Compression ratios conflate frequency skew with sequential redundancy; we mitigate this with the unigram-shuffled baseline, but residual confounding is possible. The PCA-based d_{95} reflects variance structure of a frequency matrix, not necessarily geometric manifold dimensionality; we report it as an exploratory estimate.

Internal validity. The most serious internal threat—that boilerplate drives the findings—is addressed directly in Section 5.3: full stripping changes all metrics by $< 3.2\%$. Extraction correctness was independently verified with zero mismatches across five binaries. We analyse opcodes without operands; operand-aware variants confirm that the entropy-rate gap persists (Section 5), but Zipf exponents would change with a larger vocabulary.

External validity. All binaries are x86-64 ELF on GNU/Linux. Whether the manifold hypothesis holds for ARM64, RISC-V, or WebAssembly is an open question. The three independent corpora (Section 5) demonstrate consistency within a single ISA and OS, but cross-platform replication is needed. The expanded compiler matrix (100 binaries, gcc/clang, five optimisation levels) shows compiler invariance, but other compilers (MSVC, ICC) and source languages are untested.

Statistical conclusion validity. The corpus contains $N = 15$ binaries for the primary analysis—a convenience sample, not a random draw from a well-defined population. Bootstrap CIs measure within-corpus stability, not population uncertainty. Instruction tokens within a binary are not independent; corpus-level claims (Zipf α, entropy rates) effectively have $N = 15$ independent observations, not millions. Subsampling down to $N = 4$ preserves directional conclusions but widens CIs substantially. Multiple metrics are reported without formal correction for multiple comparisons.

7 Conclusions and Future Work

We have presented *OpCode-Stats*, an open-source toolkit for statistical analysis of compiled instruction sequences through the lens of computational biology. Across a 15-binary smoke corpus (2.91 M instructions), three independent validation corpora, and a 100-binary compiler matrix, the toolkit shows that opcode frequencies are heavily concentrated (MLE $\hat{\alpha} \approx 1.4$; 95% CI [1.36, 1.53]), entropy rates decrease by 1.12 bits from $n = 1$ to $n = 5$ (vs. 0.30 bits for a shuffled baseline, confirming genuine sequential dependencies), binaries compress to 22% of raw token volume vs. 36% for unigram-shuffled sequences and 67% for random sequences, 7,623 unique k-mer motifs capture compiler-generated idioms, the corpus manifold occupies only $d_{95} = 8$ of 200 bigram dimensions, and a bigram language model achieves perplexity 8.5—within 0.07 bits of the information-theoretic lower bound.

Systematic validation (Section 5) demonstrates that these findings are not artefacts of compiler boilerplate: aggressive stripping of prologues, epilogues, and `endbr64` guards changes all metrics by at most 3.2%. The findings replicate across three disjoint corpora, and compiler identity explains less than 3% of metric variance. Of the ten main empirical claims, eight are rated strong, one (code clones) requires qualification regarding toolchain vs. application clones, and one (categorical clustering by binary type) is unsupported at the effect sizes observed.

These findings support a *thin structured manifold* hypothesis: compiled programs are concentrated in a tiny, highly regular corner of the space of all possible opcode sequences, shaped by hardware conventions, compiler passes, and algorithmic patterns. This structure explains why language models and similarity search are effective on binary code.

Future work. Several directions follow naturally from this work:

1. **Multi-ISA comparison.** Apply the same pipeline to ARM64, RISC-V, WASM, and JVM bytecode to test whether Zipf exponents and manifold dimensionality are architecture-independent.

2. **Operand-level analysis.** Extend opcode sequences to include register classes and immediate ranges, quantifying how operand information changes entropy and manifold structure.

3. **Compiler fingerprinting.** Use clustering and motif signatures to distinguish GCC from Clang from MSVC, and to infer optimisation levels ($-O0$ through $-O3$), building on the category purity results in Section 4.

4. **Cross-language manifold comparison.** Analyse binaries compiled from C, C++, Rust, Go, and Zig to determine whether source language leaves detectable fingerprints in instruction-level statistics.

5. **Neural language model evaluation.** Extend the n-gram perplexity baseline (Section 4.10) to Transformer-based models, quantifying how much additional structure beyond trigrams a neural model captures and whether attention patterns align with the thin-manifold axes identified here.

6. **Security applications.** Leverage the thin-manifold property for anomaly detection: binaries that are outliers on the manifold (high NCD from all known-good binaries, anomalous motif

distributions) may indicate obfuscation or packing.

7. **Clone-guided vulnerability propagation.** The cross-binary clone families identified in Section 4.12 are natural candidates for tracking vulnerability propagation: a known-bad function that forms a cross-binary clone family implies the vulnerability may be present in every binary containing that family member. Extending the clone taxonomy to include security-relevant semantics (e.g. constant-time violations, buffer-handling idioms) would transform clone detection into a lightweight binary auditing tool.

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